

Terna Engineering College

Computer Engineering Department

Program: Sem VI

Course: Cloud Computing Lab (CSL603)

Faculty: Preeti patil

PART A

(PART A: TO BE REFERRED BY STUDENTS)

Experiment No.1

A.1 Aim:

Study of NIST model of cloud computing

A.2 Prerequisite:

Knowledge of Networking, Distributed Computing and knowledge of Software architectures.

A.3 Objective:

Objectives this experiment is to provide students an overview of the Cloud Computing, its Architecture and deployment models, service models.

A.4 Outcome: (LO1)

After successful completion of this experiment student will be able to

- understand the architecture and services of cloud computing
- advantages of cloud computing

A.5 Theory:

Cloud Computing: Cloud computing is the dynamic delivery of IT resources and capabilities as a Service over the Internet. Cloud computing encompasses any Subscription-based or pay-per-use service that, in real time over the Internet.

- **Characteristics of Cloud Computing:**

1. On-Demand Self-Service
2. Optimal Resource Utilization

3. Minimize licensing new software
4. Reduce capital costs
5. Improve accessibility
6. Resource Pooling
7. Rapid Elasticity

Benefits of cloud:

1. Cost
2. Superior Infrastructure and Support
3. Scalability and flexibility
4. Reliability

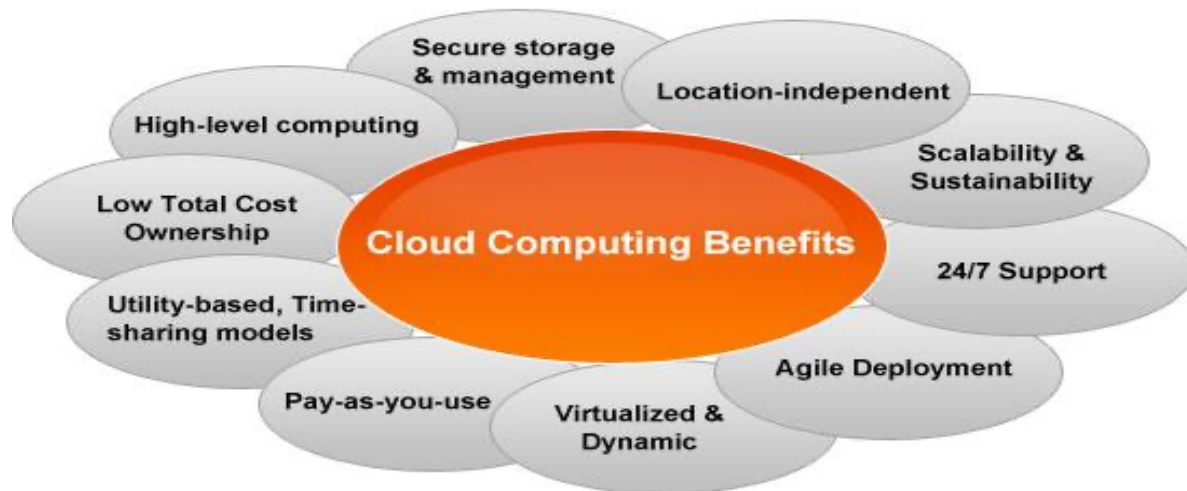


Figure 1- Benefits of Cloud Computing

Types of Cloud:

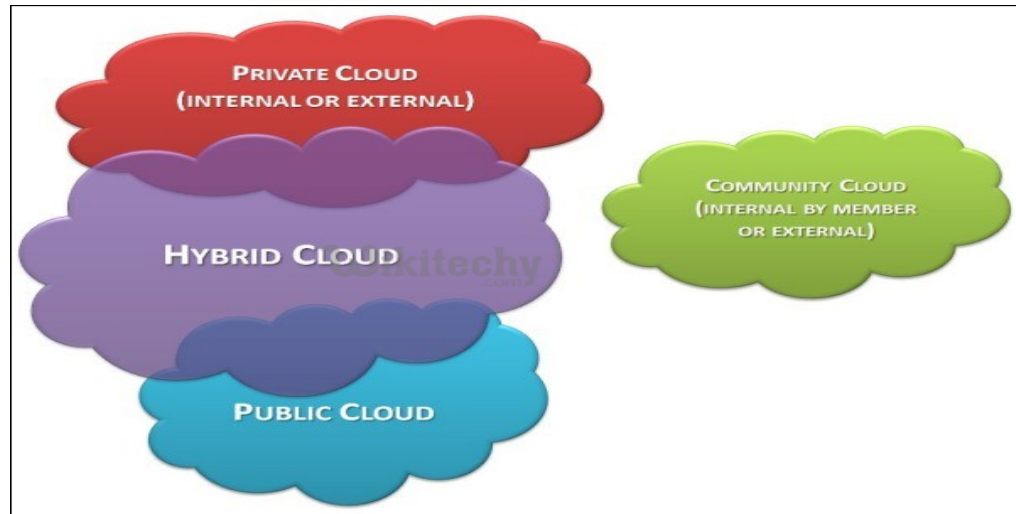
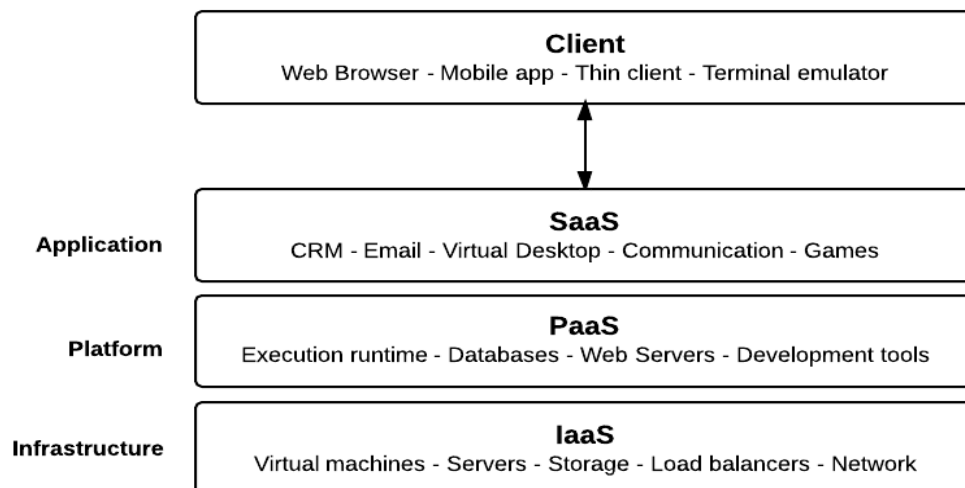


Figure 2- Types of Cloud

Service Models:

1. **Infrastructure-as-a-service (IaaS)** - It can be defined as the use of servers, storage, and virtualization to enable utility like services.
2. **Platform-as-a-service(PaaS)** - It is a development environment where a customer can create and develop applications on a provider's computing environment.
3. **Software-as-a-service (SaaS)** - It is software provider delivers and hosts an application, without the need for the customer to house and maintain the application in its own data center.



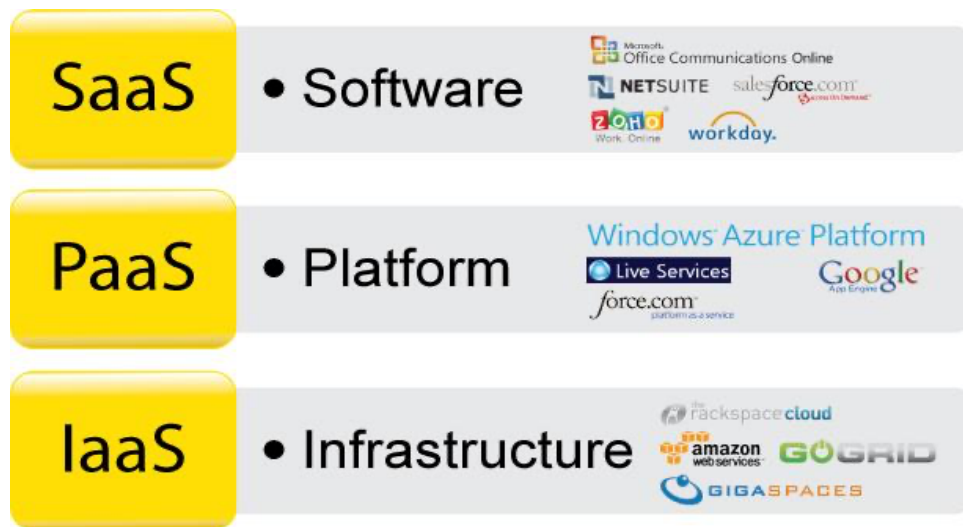


Figure 3- Service Models

NIST(National Institute of Standard and Technology)model of cloud Computing

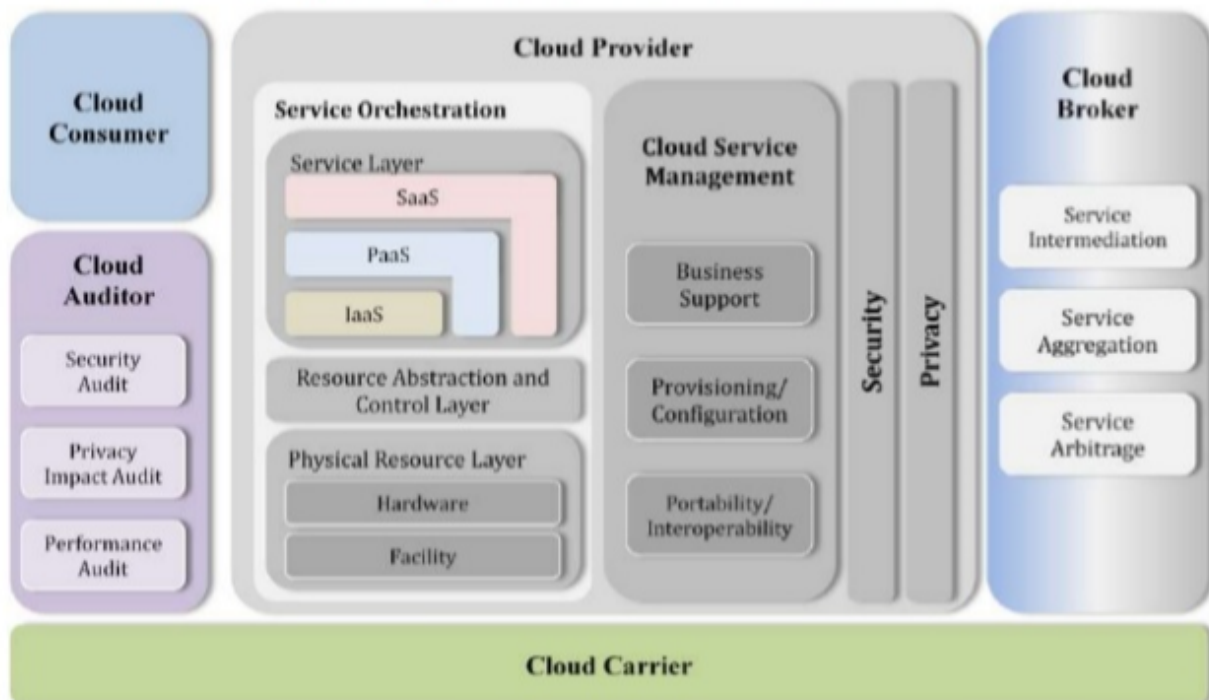


Figure 4- NIST model of Cloud Computing

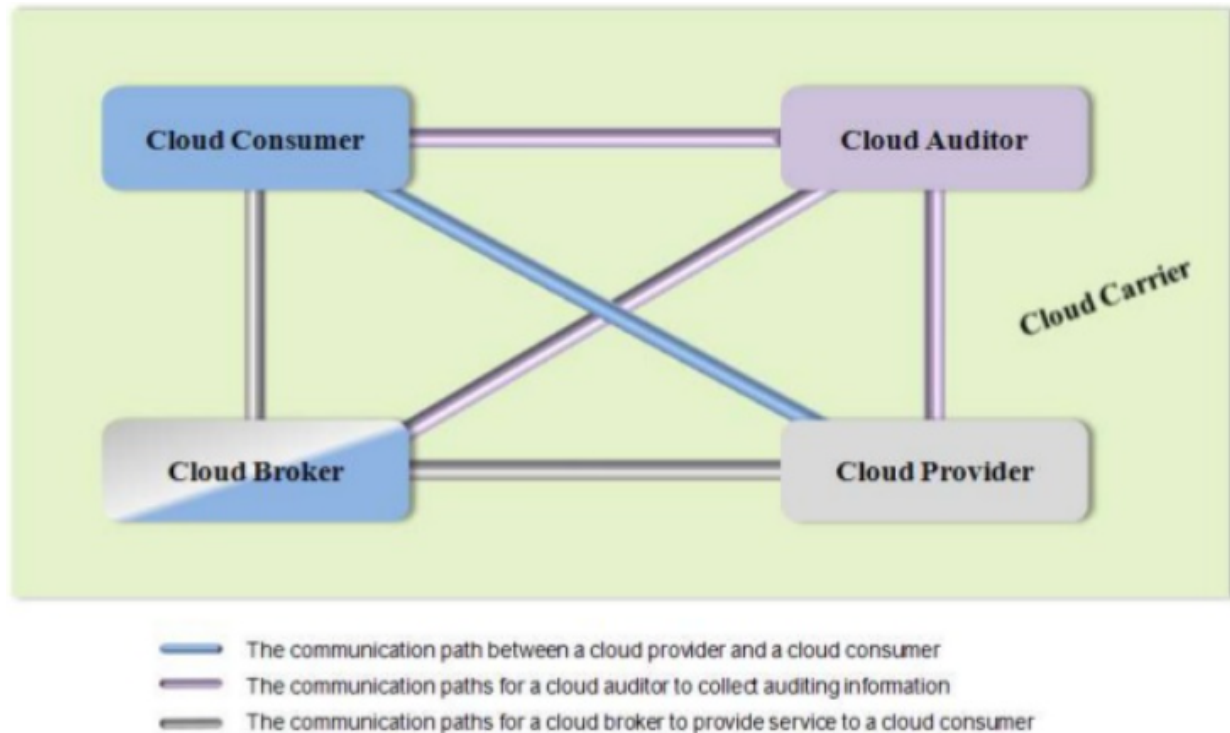


Figure 5 Interaction between actors in cloud Computing

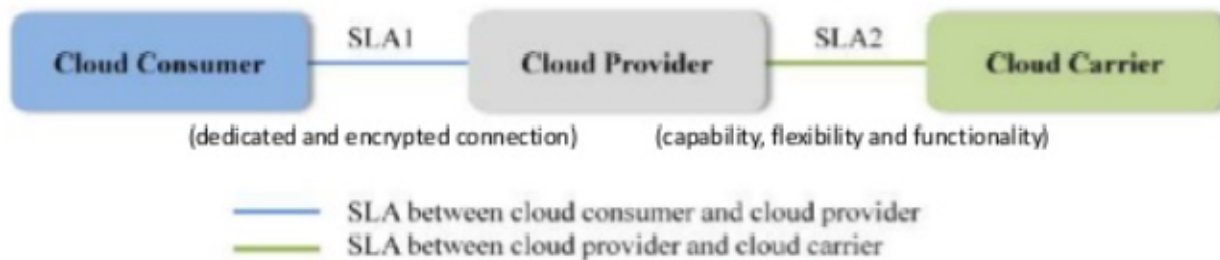


Figure 4: Usage Scenario for Cloud Carriers



Figure 6 Usage Scenario of Cloud Auditors

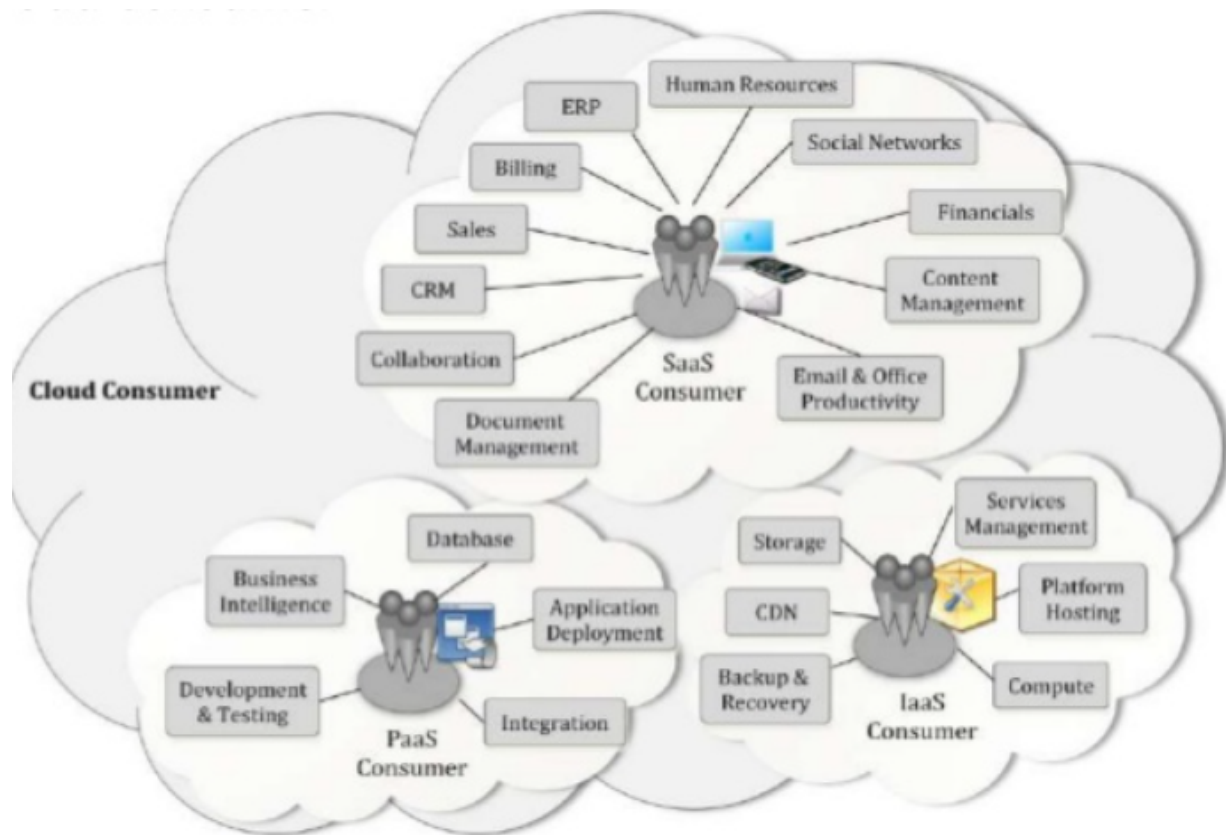


Figure 7 Services Available Over Cloud

PART B**(PART B: TO BE COMPLETED BY STUDENTS)**

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case there is no ERP access available)

Roll No. C56	Name: Quazi Md Moinuddin
Class : TE Comps C	Batch : C3
Date of Experiment: 17/01/2025	Date of Submission:5/02/2025
Grade :	

B.1 Question of Curiosity:

(To be answered by student based on the practical performed and learning/observations)

Q1: Discuss the advantages and disadvantages of Cloud Computing.

Advantages of Cloud Computing:

1. Cost-Efficiency: Lower upfront investment and reduced operational costs.
2. Scalability: Easily scale resources up or down as needed.
3. Accessibility: Access from anywhere with an internet connection.
4. Reliability: High uptime guarantees and data redundancy.
5. Automatic Updates: Regular software updates and patches handled by providers.
6. Disaster Recovery: Built-in backup and recovery options.
7. Collaboration: Real-time collaboration and file sharing.
8. Security: Advanced security features like encryption and multi-factor authentication.

Disadvantages of Cloud Computing:

1. Security and Privacy: Concerns over data breaches and control over data.
2. Downtime: Potential service outages or disruptions.
3. Limited Control: Less control over infrastructure and configurations.
4. Latency: Possible performance issues due to internet speed and distance from data centers.
5. Data Transfer Costs: Potential charges for large data egress.
6. Vendor Lock-In: Difficulty switching providers due to proprietary technologies.
7. Compliance and Legal Issues: Challenges in meeting regulatory requirements.
8. Dependence on Internet: Service availability relies on internet connectivity.

Q2: List suitable Cloud application used by top most IT companies.

Cloud applications used by top IT companies:

1. Amazon Web Services (AWS): Cloud computing, storage, machine learning (e.g., Netflix, Adobe).
2. Microsoft Azure: Cloud services, AI, IoT (e.g., LinkedIn, Samsung).
3. Google Cloud Platform (GCP): Cloud computing, big data, machine learning (e.g., Spotify, PayPal).
4. Salesforce: CRM, sales, and marketing automation (e.g., Coca-Cola, Toyota).
5. Slack: Team collaboration, messaging (e.g., IBM, Airbnb).
6. Dropbox: Cloud storage, file sharing (e.g., Spotify, Netflix).
7. Zoom: Video conferencing, meetings (e.g., Uber, Zoom itself).
8. Atlassian (Jira, Confluence): Project management, task tracking (e.g., NASA, LinkedIn).
9. Oracle Cloud: Cloud computing, databases, ERP (e.g., Zoom, FedEx).
10. IBM Cloud: AI, data storage, machine learning (e.g., Verizon, Bank of America).

11. ServiceNow: IT service management, automation (e.g., Microsoft, Accenture).

12. GitHub: Code repository, version control (e.g., Microsoft, Google).

Q3: List all Cloud Providers.

Major cloud providers:

1. **Amazon Web Services (AWS):** Comprehensive cloud services (computing, storage, AI, etc.).
2. **Microsoft Azure:** Cloud services, AI, DevOps, security, and more.
3. **Google Cloud Platform (GCP):** Cloud computing, big data, and machine learning.
4. **IBM Cloud:** Cloud infrastructure, AI, blockchain, IoT.
5. **Oracle Cloud:** Cloud computing, databases, ERP, CRM.
6. **Alibaba Cloud:** Cloud computing, AI, big data, and security.
7. **Salesforce:** Cloud CRM, sales automation, marketing.
8. **DigitalOcean:** Cloud computing, virtual private servers, Kubernetes.
9. **VMware Cloud:** Virtualization and cloud management.
10. **Tencent Cloud:** Cloud services, networking, AI, and storage.
11. **Huawei Cloud:** Cloud infrastructure, AI, IoT.
12. **SAP Cloud:** Cloud-based ERP, analytics.

Q4: List Various SaaS Services

SaaS platforms cater to diverse needs across various domains:

1. Communication & Collaboration: Google Workspace, Microsoft 365, Zoom.
2. CRM: Salesforce, HubSpot, Zoho CRM.
3. Marketing & Analytics: Mailchimp, Google Analytics, SEMrush.
4. E-commerce: Shopify, WooCommerce, Wix.

5. ERP: NetSuite, SAP Business ByDesign.
6. HRMS: BambooHR, Workday, Gusto.
7. File Storage: Google Drive, Dropbox, OneDrive.
8. Finance & Accounting: QuickBooks, Xero, FreshBooks.
9. Design & Creativity: Canva, Adobe Creative Cloud, Figma.
10. DevOps: GitHub, GitLab, Jenkins.
11. Learning Management: Moodle, Blackboard, TalentLMS.
12. Cybersecurity: Okta, Norton 360, LastPass.
13. Business Intelligence: Tableau, Power BI, Looker.
14. Virtual Events: Hopin, GoToWebinar, Airmeet.
15. Health & Fitness: MyFitnessPal, Headspace, Peloton

Q5: List Various Paas Services

Popular PaaS Providers:

1. Google Cloud Platform (App Engine)
2. Microsoft Azure (App Services)
3. Amazon Web Services (Elastic Beanstalk)
4. Heroku
5. Red Hat OpenShift
6. IBM Cloud Foundry
7. Oracle Cloud PaaS
8. Salesforce Platform
9. OutSystems (Low-code PaaS)
10. Mendix (Low-code PaaS)

Q6:List Various IaaS Services

Popular IaaS Providers:

1. **Amazon Web Services (AWS):** EC2, S3, VPC.
2. **Microsoft Azure:** Virtual Machines, Blob Storage.
3. **Google Cloud Platform (GCP):** Compute Engine, Cloud Storage.
4. **IBM Cloud Infrastructure:** Virtual Servers, Block Storage.
5. **Oracle Cloud Infrastructure (OCI):** Compute, Autonomous Database.
6. **Alibaba Cloud:** Elastic Compute Service (ECS), Object Storage.
7. **DigitalOcean:** Droplets, Block Storage.
8. **Linode:** Cloud Hosting & Virtual Servers.
9. **VMware vCloud Air:** Hybrid Cloud Infrastructure.
10. **Rackspace:** Managed Cloud Hosting.

B.2 Conclusion:

The NIST model provides a comprehensive and standardized framework to understand cloud computing. It emphasizes flexibility, scalability, and efficiency while addressing diverse user needs through different service and deployment models. This model forms the foundation for evaluating and implementing cloud solutions across industries.